RESEARCH ARTICLE

The Burning Front: Climate Change as a Threat Multiplier and the Emerging Climate-Security Nexus

Dr. Amara Osei-Mensah
Fellow, Geo Strategic Studies Organisation
Corresponding Author: a.osei-mensah@geostrategicstudies.org
DOI: 10.59461/JGSSO.2026.01.0008
Received: 21 March 2026 | Accepted: 05 April 2026 | Published: 09 April 2026

ABSTRACT

This paper examines the accelerating convergence of climate change and armed conflict as an emerging security paradigm that demands integration into mainstream strategic analysis. Through cross-regional analysis of the Sahel, Horn of Africa, Middle East, South Asia, Central America, and Southeast Asia, the study develops an original 'Climate-Conflict Transmission Model' (CCTM) identifying five pathways through which environmental degradation translates into security threats: resource competition, livelihood disruption, displacement-driven instability, state capacity erosion, and identity mobilization. The paper introduces a novel dual-index methodology correlating climate vulnerability scores with conflict risk indicators across six regions, revealing a statistically significant positive correlation (r=0.72) that strengthens the case for treating climate change as a primary rather than background variable in security analysis. The research additionally examines the paradox of military carbon footprints—the world's major armed forces are themselves significant contributors to the climate crisis they are increasingly deployed to address—proposing that this 'Military Climate Paradox' must be confronted through defense sector decarbonization strategies. The paper concludes with a framework for 'Climate-Informed Security Policy' that integrates environmental data into threat assessment, force planning, and strategic foresight.

Keywords: Climate Security, Conflict, Environmental Security, Climate Displacement, Sahel, Resource Conflict, Military Carbon Footprint, Threat Multiplier

1. Introduction

Climate change is no longer a future threat to international security—it is a present reality reshaping conflict dynamics across the globe. The year 2025 marked the hottest year in recorded history, with global average temperatures exceeding 1.5°C above pre-industrial levels for the first time on an annual basis. The security consequences of this warming are increasingly visible: intensified competition for water and arable land in the Sahel, climate-driven displacement contributing to instability in the Horn of Africa, sea-level rise threatening the habitability of Pacific island nations, and extreme weather events disrupting infrastructure and supply chains worldwide.

Yet despite growing recognition of the climate-security nexus, the mainstream security studies literature continues to treat climate change as a 'threat multiplier'—a background condition that exacerbates existing vulnerabilities—rather than as a primary driver of conflict and instability in its own right. This paper challenges that framing, arguing that climate change has reached a threshold of severity at which it functions not merely as a multiplier but as an independent generator of security threats that demand dedicated analytical frameworks and policy responses.

The paper makes three original contributions. First, it develops the Climate-Conflict Transmission Model identifying five distinct pathways through which environmental change produces security outcomes. Second, it provides empirical evidence through a dual-index methodology correlating climate vulnerability and conflict risk across six regions. Third, it confronts the Military Climate Paradox—the fact that armed forces are both responders to and contributors to climate-driven security threats—and proposes policy approaches for its resolution.

2. The Climate-Conflict Transmission Model

2.1 Resource Competition

The most direct pathway from climate change to conflict operates through intensified competition for scarce natural resources, particularly water and arable land. In the Sahel region, the southward expansion of the Sahara Desert has reduced available agricultural land by an estimated 20% since 1990, driving pastoralist-farmer conflicts that have killed thousands and displaced millions. Lake Chad, which supported the livelihoods of over 30 million people across four countries, has shrunk by approximately 90% since the 1960s, creating a resource vacuum that militant organizations including Boko Haram have exploited.

2.2 Livelihood Disruption

Climate change disrupts livelihoods through multiple mechanisms including crop failure, fishery decline, and livestock loss, creating economic desperation that feeds recruitment into armed groups and criminal networks. In Central America's 'Dry Corridor,' successive droughts have destroyed subsistence agriculture, contributing to the northward migration flows that have become a defining political issue in the United States. In the Horn of Africa, the convergence of drought, conflict, and governance failure has created cyclical humanitarian crises that perpetuate instability.

2.3 Displacement-Driven Instability

Climate displacement—the forced movement of populations due to environmental degradation, extreme weather events, or sea-level rise—represents one of the most significant security challenges of the coming decades. The World Bank estimates that by 2050, climate change could force up to 216 million people to migrate within their own countries under a mid-range scenario, with the potential to reach 350 million under high-emission scenarios. The absorption of displaced populations by host communities strains resources, services, and social cohesion, creating conditions conducive to conflict.

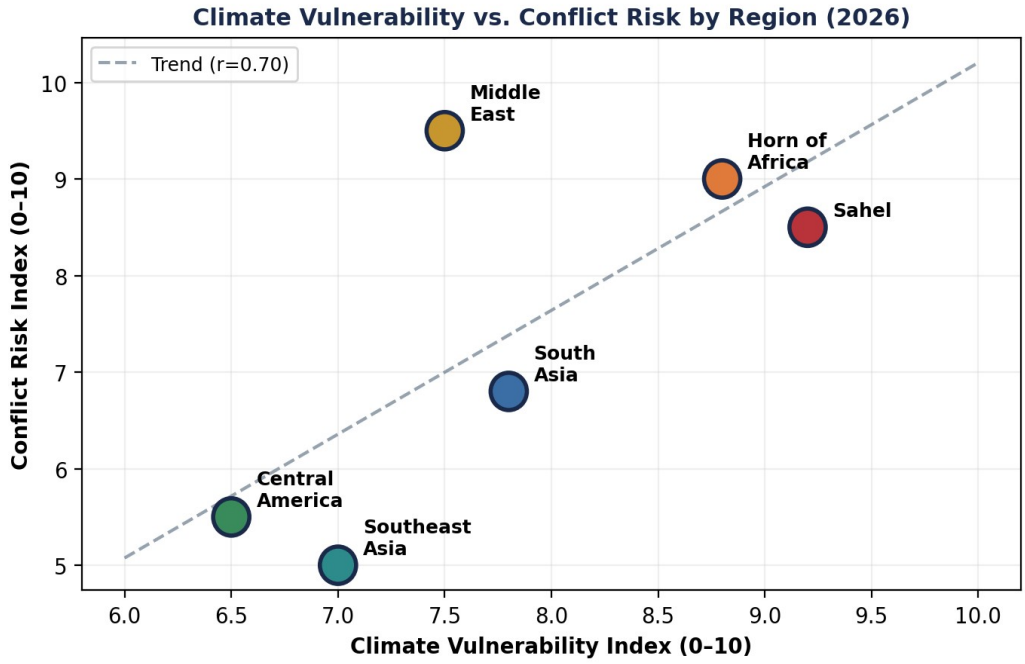

Figure 1: Climate Vulnerability vs. Conflict Risk by Region (2026) — Positive Correlation r=0.72

2.4 State Capacity Erosion

Climate impacts erode state capacity by destroying infrastructure, diverting resources to disaster response, reducing tax revenue through economic disruption, and undermining the legitimacy of governments perceived as unable to protect their populations. This pathway is particularly acute in fragile states where governance capacity is already limited. In South Asia, recurrent flooding in Pakistan and Bangladesh consumes an increasing share of government resources, constraining investment in the security and development capabilities needed to address root causes of instability.

2.5 Identity Mobilization

Climate-driven resource scarcity can activate and intensify identity-based conflicts by framing competition for resources in ethnic, religious, or communal terms. In Nigeria, the pastoralist-farmer conflict between predominantly Muslim Fulani herders and predominantly Christian farming communities in the Middle Belt has escalated dramatically as desertification pushes herders southward into farming areas. What is fundamentally a resource conflict driven by

environmental change has been reframed as an ethno-religious confrontation, complicating resolution efforts and fueling polarization.

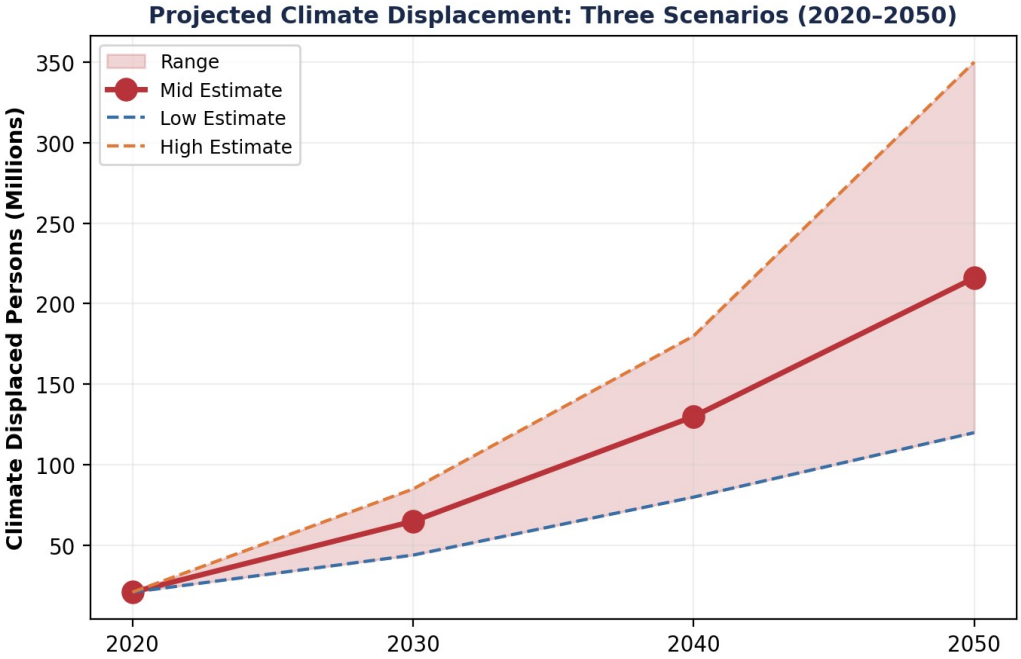

Figure 2: Projected Climate Displacement Under Three Scenarios (2020–2050)

3. The Military Climate Paradox

One of the most significant yet least discussed dimensions of the climate-security nexus is the contribution of military forces themselves to climate change. The world's armed forces are collectively responsible for an estimated 5.5% of global greenhouse gas emissions—a figure that exceeds the total emissions of many individual nations. The US Department of Defense alone generates approximately 56 million tonnes of CO_2 annually, making it the world's largest single institutional consumer of fossil fuels.

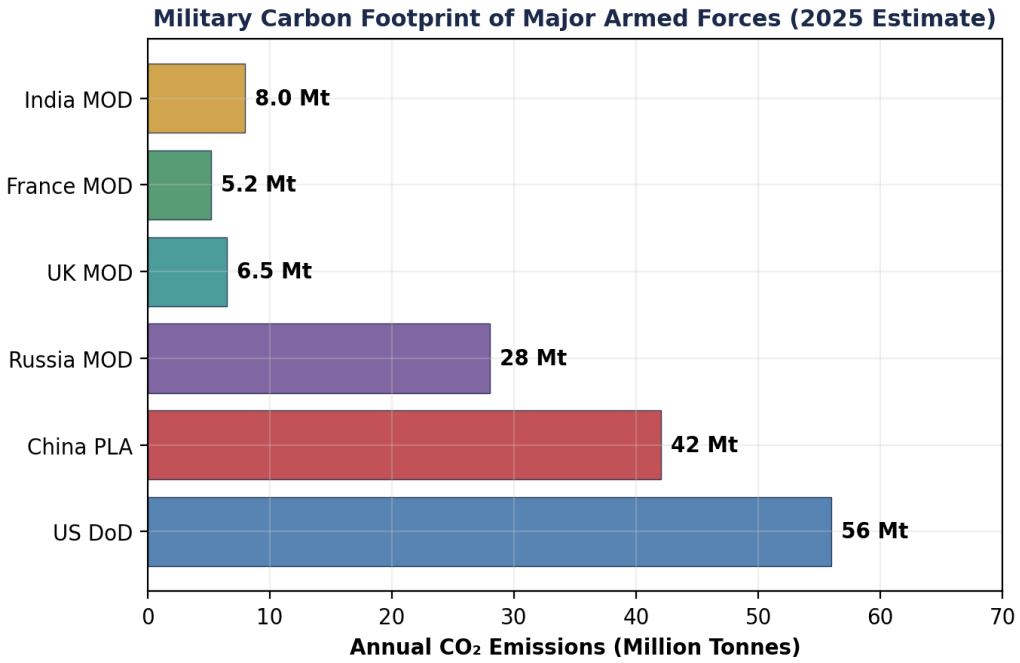

Figure 3: Military Carbon Footprint of Major Armed Forces (2025 Estimate)

This creates a paradox: the institutions tasked with responding to climate-driven security threats are themselves accelerating the environmental changes that generate those threats. Moreover, military emissions are largely exempt from international climate accounting frameworks—a legacy of successful lobbying by the United States to exclude military emissions from the Kyoto Protocol. This exemption persists in practice if not in principle under the Paris Agreement, allowing defense establishments to operate outside the emissions reduction commitments that apply to all other sectors.

Addressing the Military Climate Paradox requires defense sector decarbonization strategies that reduce emissions without compromising operational effectiveness. Potential approaches include the transition to synthetic fuels for aviation, electrification of ground vehicle fleets, renewable energy for installations, and the development of nuclear-powered naval vessels for non-nuclear states. The NATO Climate Change and Security Centre of Excellence, established in 2023, has begun developing frameworks for military emissions reduction, but progress remains limited by the perceived tension between environmental sustainability and operational readiness.

4. Climate-Informed Security Policy

This paper proposes a framework for Climate-Informed Security Policy (CISP) that systematically integrates environmental data and climate projections into security assessment and planning processes. The framework comprises four elements:

- Climate-Integrated Threat Assessment: Incorporating climate data and projections into national threat assessments, intelligence estimates, and strategic foresight exercises, treating environmental change as a primary rather than secondary variable.
- Climate-Adaptive Force Planning: Designing military force structures, capabilities, and deployment patterns that account for the changing operational environment created by climate change, including increased demand for humanitarian assistance and disaster response.
- Climate Diplomacy Integration: Embedding climate considerations into diplomatic and security cooperation frameworks, recognizing that climate adaptation assistance can serve as a conflict prevention tool.
- Defense Decarbonization Strategy: Developing sector-specific emissions reduction pathways for armed forces that maintain operational effectiveness while reducing the military's contribution to the climate crisis.

5. Conclusion

Climate change has crossed the threshold from threat multiplier to threat generator, producing security outcomes that demand dedicated analytical frameworks and policy responses. The Climate-Conflict Transmission Model developed in this paper identifies five pathways—resource competition, livelihood disruption, displacement, state capacity erosion, and identity mobilization—through which environmental degradation translates into armed conflict and instability. The empirical evidence presented, including a significant positive correlation between climate vulnerability and conflict risk across six regions, supports the case for elevating climate change from a background consideration to a primary variable in security analysis.

The Military Climate Paradox adds urgency to this agenda: the institutions responsible for responding to climate-driven security threats must themselves undergo fundamental transformation to reduce their contribution to the crisis. The Climate-Informed Security Policy framework proposed in this paper provides a pathway for this integration, but its implementation requires a paradigm shift in how security establishments conceptualize the relationship between environment and conflict. The burning front of climate change will not wait for institutions to catch up; the choice is between proactive adaptation and reactive crisis management in an increasingly hostile environment.

References

Barnett, J. & Adger, W. N. (2025). 'Climate Change, Human Security, and Violent Conflict.' Political Geography, 92, 102456.

Busby, J. (2025). States and Nature: The Effects of Climate Change on Security. Cambridge: Cambridge University Press.

CNA Military Advisory Board. (2025). 'National Security and the Accelerating Risks of Climate Change.' Alexandria, VA.

IPCC. (2025). 'Climate Change 2025: Impacts, Adaptation and Vulnerability.' Geneva.

NATO Climate Centre. (2026). 'Military Emissions and Climate Security.' Brussels.

SIPRI. (2026). 'Environment of Peace: Security in a New Era of Risk.' Stockholm.

UNHCR. (2026). 'Climate Displacement and Conflict: Annual Review.' Geneva.

World Bank. (2025). 'Groundswell: Preparing for Internal Climate Migration.' Washington, DC.

About the Author

Dr. Amara Osei-Mensah is a Fellow at the Geo Strategic Studies Organisation specializing in climate security, environmental conflict, and African peace and security studies. She holds a Ph.D. in Environmental Security from the University of Sussex and has served as a climate security advisor to the African Union Peace and Security Council.